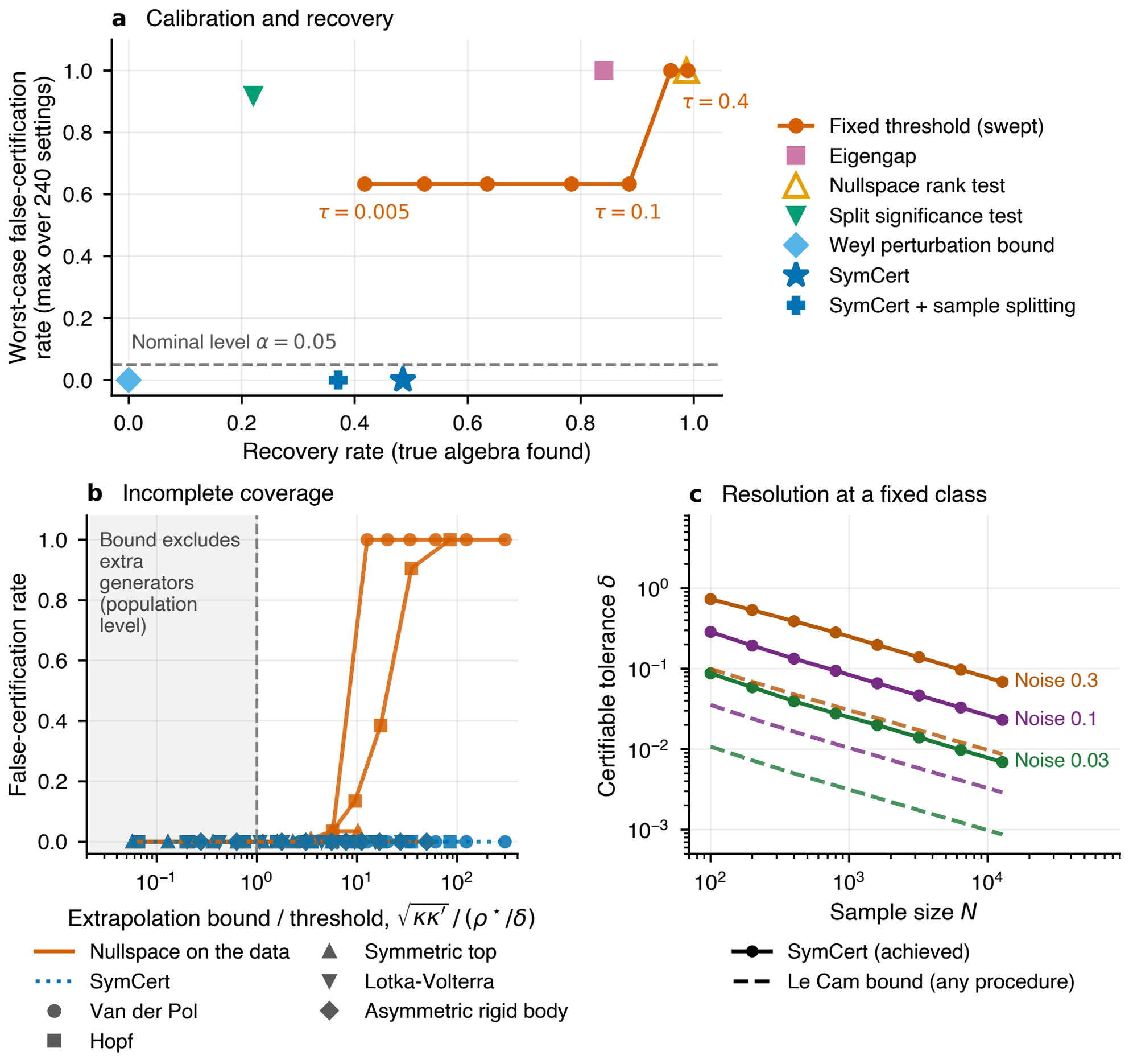
a Calibration and recovery
Worst-case false-certification rate (max over 240 settings)
Recovery rate (true algebra found)
Nominal level α = 0.05
τ = 0.005
τ = 0.1
τ = 0.4
Fixed threshold (swept)
Eigengap
Nullspace rank test
Split significance test
Weyl perturbation bound
SymCert
SymCert + sample splitting
b Incomplete coverage
False-certification rate
Bound excludes extra generators (population level)
Extrapolation bound / threshold, √(κκ′) / (ρ* /δ)
Nullspace on the data
SymCert
Van der Pol
Hopf
Symmetric top
Lotka-Volterra
Asymmetric rigid body
c Resolution at a fixed class
Certifiable tolerance δ
Sample size N
Noise 0.3
Noise 0.1
Noise 0.03
SymCert (achieved)
Le Cam bound (any procedure)